

Isaac Mubiru

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PROFESSIONAL SUMMARY

Senior Backend Engineer with 5+ years building high-volume data processing systems and real-time pipelines using Node.js, TypeScript, and cloud platforms. Led zero-downtime migration of 20+ production applications from MongoDB to Firestore, designing event-driven architectures with Apache Kafka and message queues. Proven track record working autonomously in remote-first environments, delivering infrastructure improvements that reduced costs by \$5,000/month. Strong foundation in Docker, Kubernetes, and AWS with experience building agentic AI integrations for production workflows.

CORE COMPETENCIES

TypeScript / Node.js

High-Volume Data Systems

Apache Kafka

Real-Time Data Pipelines

Docker & Kubernetes

AWS / GCP

MongoDB / PostgreSQL

Message Queues (Pub/Sub, gRPC)

WORK EXPERIENCE

MTailor

September 2022 – Present

Software Engineer — US Remote

- Led end-to-end migration of 20+ production applications from Parse/MongoDB to Firebase/GCP with zero downtime, reporting directly to the CTO
- Implemented **real-time two-way sync pipeline** processing high-volume data streams between MongoDB and Firestore using Node.js and Google Pub/Sub
- Built 3D visualization feature using video overlay and ffmpeg library, **increasing buyer conversion**
- Migrated file storage from Amazon S3 to GCS via Python script, ensuring data integrity across cloud platforms
- Saved \$5,000/month** in infrastructure costs by migrating all services from AWS to GCP
- Implemented Express Shipping feature, generating an additional \$40 revenue per order

MTailor

March 2022 – July 2022

Contractor — US Remote

- Built migration script to process high-volume data from MongoDB to Firestore with complex transformation logic
- Wrote comprehensive documentation for engineers on working with Firebase SDKs (Node.js and Python)
- Trained Ops team on new Firebase Dashboard and backup procedures

Dr Wealth

August 2021 – January 2022

Software Engineer — Singapore Remote

- Optimized PWA delivering real-time stock market prices from Firebase backend
- Built ad-hoc jobs querying **2+ million Firestore records** to keep customer prices current from Morningstar APIs
- Extended backend APIs using Firebase Cloud Functions and Express.js hosted on Heroku

Mind2matter.co

January 2022 – July 2022

Software Engineer — US Remote

- Built backends for DeFi applications using Web3 and Node.js
- Delivered React UIs from Figma designs under tight timelines, reporting directly to CTO

CodeBits

January 2020 – July 2021

Software Engineer — Uganda Remote

- Led team of 4 developers building legal tech systems for NGOs in Uganda
- Architected **event-driven microservices backend** using Apache Kafka for distributed message processing, Docker, and Kubernetes
- Built USSD service for legal aid providers communicating with Node.js backend via gRPC
- Developed cross-platform mobile app using React Native and Expo with Firebase push notifications

PROJECTS

Healthcare WhatsApp Chatbot AI/ML

Integrated OpenAI + Google Vision for agentic prescription OCR workflow via WhatsApp Business API. Automated prescription handling with AI-powered decision making.

Node.js, OpenAI API, Google Vision, WhatsApp Business API

Marketplace Recommendation Engine Data Systems

Built collaborative + content-based filtering engine in MongoDB for service provider matching.

MongoDB, Node.js, Machine Learning algorithms

Real-Time Pub/Sub Sync Pipeline Infrastructure

Built live MongoDB to Firestore bidirectional sync pipeline supporting parallel production traffic during migration with zero data loss.

Node.js, Google Pub/Sub, MongoDB Change Streams, Firestore

EDUCATION

Bachelor of Science in Computer Engineering — Makerere University

2018 – 2022

Kampala, Uganda

SKILLS

Languages: TypeScript, JavaScript, Python, Go, Rust **Backend:** Node.js, Express.js, Firebase Cloud Functions

Data: MongoDB, PostgreSQL, Firestore, Redis **Message Queues:** Apache Kafka, Google Pub/Sub, NATS, gRPC

Cloud: GCP, AWS (S3, EC2, EBS), Firebase **DevOps:** Docker, Kubernetes, Git, Linux **Frontend:** React, React Native, HTML, CSS, Tailwind